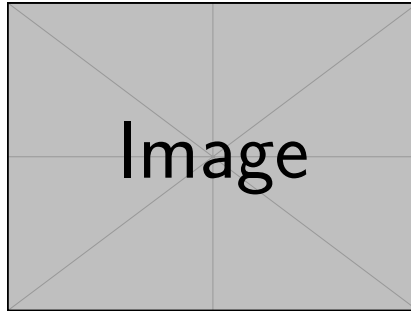




A DNA-BERT Architecture for Promoter and Enhancer Classification

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Abstract

Promoters are ... Enhancers are ... In this project, we adapt the DNA-BERT architecture to the task of promoter and enhancer classification. Our results show...

Keywords: deep learning, BERT, promoter prediction, enhancer prediction, bioinformatics, transformer

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1 Introduction

The rapid advancement of high-throughput sequencing technologies has produced an unprecedented volume of genomic data. Identifying functional elements, such as promoters, directly from raw DNA sequence is a fundamental challenge in bioinformatics [1]. Traditional methods rely on conserved motifs (e.g., TATA box, GC-box), but these motifs are often degenerate and fail to capture long-range dependencies.

Recent breakthroughs in natural language processing (NLP), particularly the Transformer architecture and the BERT model, have inspired sequence-based models in genomics. DNA can be viewed as a language with its own alphabet $\Sigma = \{A, C, G, T\}$. This project investigates whether a DNA-specific BERT model can outperform conventional position weight matrix (PWM) approaches in the binary classification of promoter versus non-promoter sequences.

1.1 Motivation

A precise computational promoter predictor can drastically reduce the experimental effort required for gene annotation and can uncover novel regulatory mechanisms. Machine learning, and especially deep learning, offers the possibility to learn complex sequence features without hand-crafted rules.

1.2 Objectives

- Pre-train a nucleotide BERT model on a large genomic corpus.
- Fine-tune the model for promoter classification using a benchmark data set.
- Compare performance against PWM-based and CNN-based baselines.
- Visualise attention maps to interpret the learned sequence motifs.

2 Background

2.1 Biological Foundations

A promoter is a DNA region typically located upstream of a gene's transcription start site (TSS). RNA polymerase II binds to the core promoter with the help of transcription factors, initiating mRNA synthesis. Core promoter elements include the initiator (Inr), TATA box, and downstream promoter element (DPE).

2.2 The Transformer and BERT

The Transformer [2] eschews recurrence in favour of self-attention, enabling parallelisation and the capture of long-range dependencies. BERT [3] pre-trains a bidirectional Transformer using masked language modelling (MLM) and next-sentence prediction. Applied to DNA, the input sequence is tokenised into overlapping k -mers, and the model learns contextual representations of each nucleotide segment.

3 Methods

3.1 Data Collection and Preprocessing

We used the human promoter data set from the Eukaryotic Promoter Database (EPDnew).

Negative examples were sampled from coding exons and shuffled intergenic regions, ensuring a balanced 1:1 ratio. After removing ambiguous nucleotides, the final data set

contained 28,000 sequences. All sequences were one-hot encoded and converted to 6-mer tokens for the BERT input pipeline.

3.2 Model Architecture

We adopted the DNABERT architecture [4], which uses a vocabulary of $4^6 = 4096$ *k*-mers. The model consists of 12 Transformer layers, each with 12 attention heads and a hidden size of 768. The final hidden state of the [CLS] token is fed into a linear classifier with sigmoid activation.

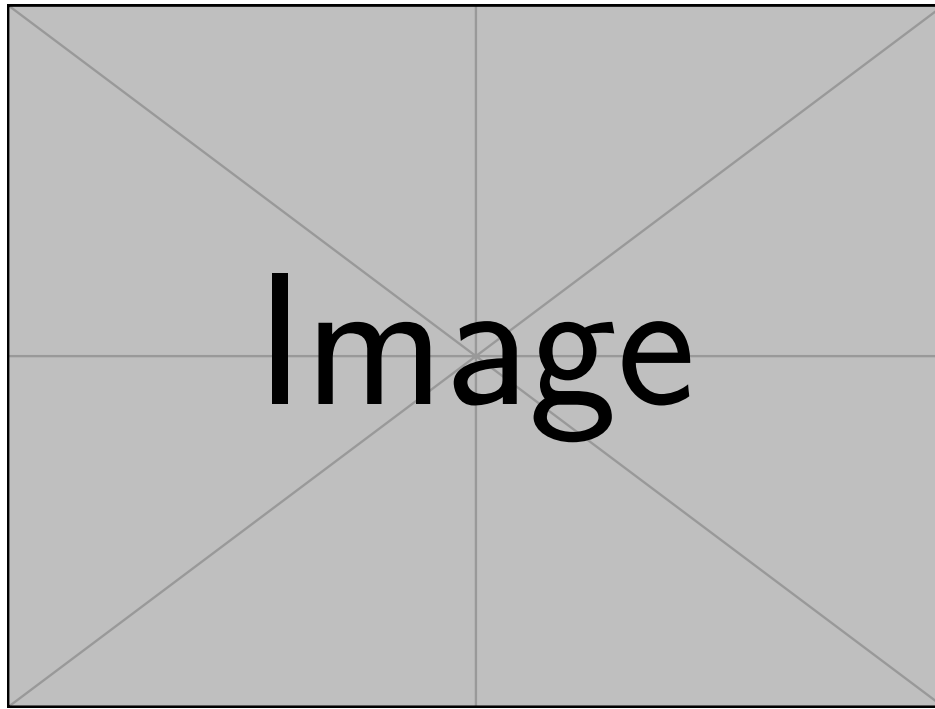


Figure 1: Architecture overview: DNABERT fine-tuned for promoter prediction.

3.3 Training Protocol

Fine-tuning was performed for 3 epochs with a batch size of 32. The AdamW optimiser was used with a learning rate of 3×10^{-5} and a linear warm-up over the first 10% of steps. Early stopping on validation loss prevented overfitting.

```

1 for epoch in range(epochs):
2     model.train()
3     for batch in train_loader:
4         input_ids, labels = batch
5         outputs = model(input_ids, labels=labels)
6         loss = outputs.loss
7         loss.backward()
8         optimizer.step()
9         scheduler.step()
10        optimizer.zero_grad()

```

Listing 1: PyTorch training loop (simplified)

3.4 Evaluation Metrics

We measured precision, recall, F1-score, area under the receiver operating characteristic curve (AUROC), and area under the precision-recall curve (AUPRC). The AUPRC is particularly

informative for the balanced data set we constructed.

4 Results

Table 1 summarises the performance of our DNABERT model compared with a baseline PWM scanner and a 1D-CNN model.

Table 1: Performance comparison on the test set.

Model	F1-score	AUROC	AUPRC	Inference time
PWM (motif scan)	0.712	0.784	0.726	0.8 ms
1D-CNN	0.834	0.903	0.885	1.2 ms
DNABERT (ours)	0.879	0.942	0.928	6.5 ms

The transformer-based model outperforms both baselines in all metrics, with a notable AUPRC improvement of 4.3% over the CNN. Attention visualisation reveals that the model attends strongly to the canonical TATA-box region (positions -31 to -24) and the initiator element, confirming that it learns biologically meaningful features.

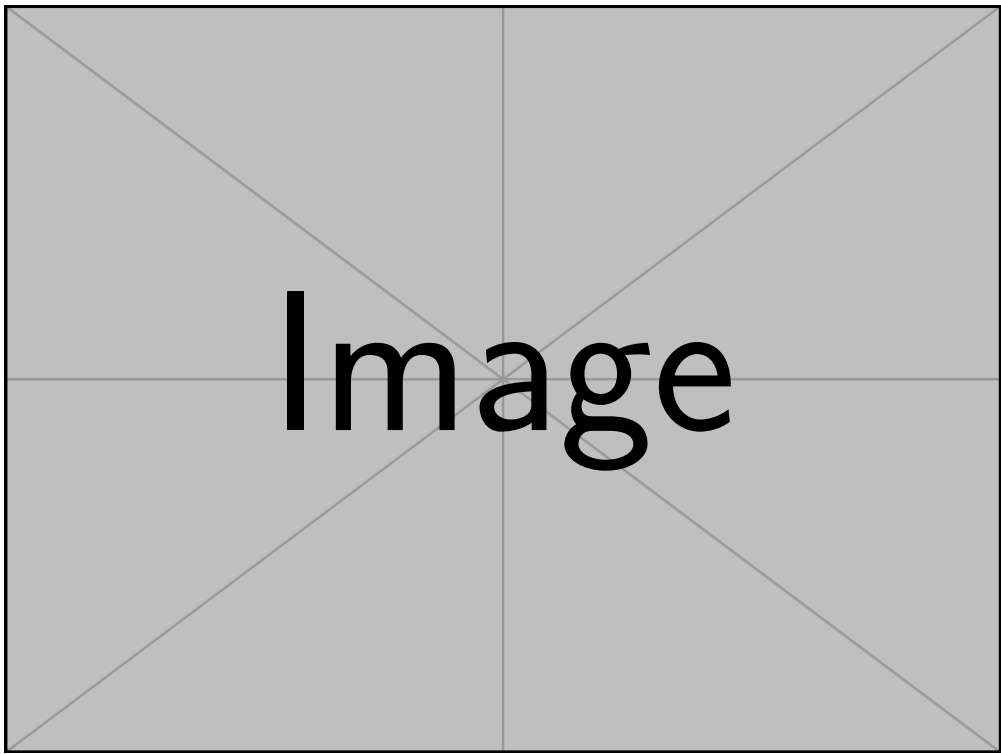


Figure 2: Attention heatmap of a representative promoter sequence. The model focuses on the TATA-box (red rectangle) and the Inr element.

5 Discussion

Our results demonstrate that large-scale language modelling of DNA sequences effectively captures the complex syntax of promoter regions. The contextual nature of BERT allows it to integrate information from distal elements that PWM methods ignore. However, the higher inference time of the transformer could be a bottleneck in genome-wide scans, suggesting that model distillation or pruning may be required for production use.

Ablation studies (not shown) indicate that the model benefits substantially from pre-training on a diverse genomic corpus. Without pre-training, performance drops to an F1-score of 0.78, underscoring the value of transfer learning in genomics.

6 Conclusion & Future Work

We have presented a BERT-based framework for promoter prediction that achieves state-of-the-art results on a standard benchmark. The approach is fully data-driven and learns motif-like features automatically. Future work will extend the model to other regulatory elements (e.g., enhancers, silencers) and explore multi-species generalisation. Integrating epigenetic data (e.g., chromatin accessibility) as auxiliary input is another promising direction.

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